UE24CS352A - MACHINE LEARNING MINI-PROJECT REPORT

Detecting and Predicting Earthquake Ground Motion Directionality Patterns Using Machine Learning Tools

Reference Study: Alan Poulos & Rodrigo Silva-Lopez (Blume Earthquake Engineering Center, Stanford University)

1. Problem Statement & Engineering Motivation

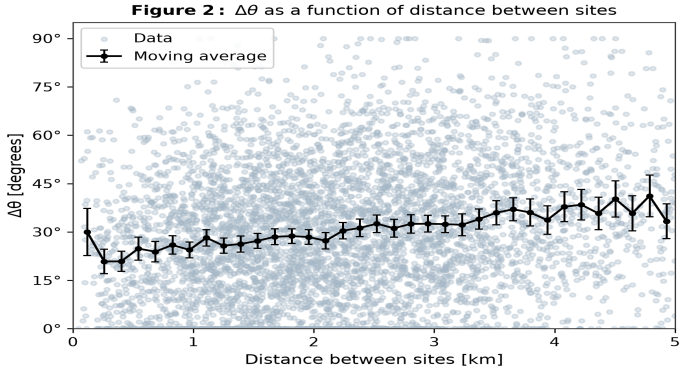
Earthquake ground shaking intensity is inherently direction-dependent (a phenomenon termed *ground motion directionality*). At any recording station, there exists a critical orientation associated with maximum spectral acceleration, standardized in structural engineering as **RotD100**. In contemporary seismic hazard analysis and regional risk assessments, orientations at different locations are conventionally assumed to follow an independent uniform distribution over [0°, 180°). However, empirical observations demonstrate that orientations between closely spaced sites exhibit strong spatial coherence due to coherent wave propagation, fault directivity, and source radiation patterns. Consequently, independent sampling underpredicts simultaneous catastrophic shaking across distributed civil infrastructure (e.g., bridges, lifelines, pipeline networks). This project develops supervised machine learning algorithms to predict the acute angular difference in direction of maximum shaking, $\Delta\theta \in$ **[0°, 90°]**, between two recording sites s_i and s_j during an earthquake E_k: $\Delta\theta_{i,j}^{k} = f(s_i, s_j, E_k)$.

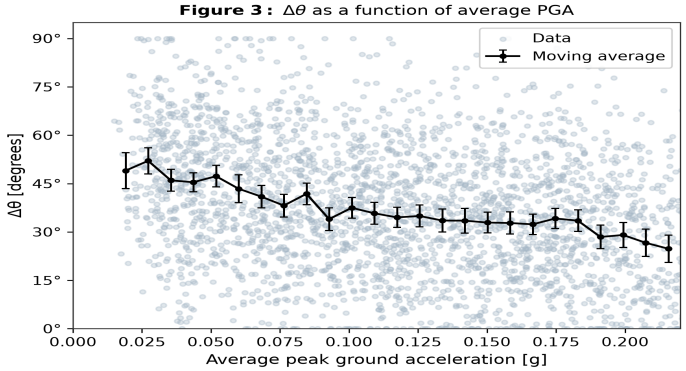
2. Dataset Characteristics & Feature Engineering

Database: The study utilizes the Pacific Earthquake Engineering Research (PEER) **NGA-West2** database containing shallow crustal records worldwide. Earthquakes with moment magnitude M_w < 5.0 are filtered out to preserve engineering significance, retaining ~5,000 ground motion records.

Pairwise Construction: To predict directionality differences, all stations recording the same seismic event within an inter-site distance threshold $d \leq 5.0$ km are paired, synthesizing **5,300 pairwise instances**. For each pair, the target variable $\Delta\theta$ was computed from recorded acceleration waveforms as $\Delta\theta = \min(|\theta_i - \theta_j|, 180° - |\theta_i - \theta_j|)$.

Initial Feature Pool: 71 seismological, geological, and instrument attributes were extracted: earthquake source parameters (M_w, focal mechanism, strike, dip, rake angles, rupture length, width, area, hypocenter depth, ZTOR), dual-site distance metrics (R_{epi}, R_{hyp}, R_{jb}, R_{rup}), geotechnical soil properties (shear wave velocity V_{s30}, NEHRP site classes B–E), and shaking intensity metrics (PGA, PGV, PGD). Numerical attributes were standardized to zero mean and unit variance, and categorical attributes were one-hot encoded.





Figures 1 & 2: *Empirical trends of angular difference Δθ. Left: Δθ increases monotonically with inter-site distance d up to 5 km. Right: Δθ decreases with higher average Peak Ground Acceleration (PGA) due to enhanced source directivity alignment. Error bars denote 95% CI.*

3. Stepwise Backward-Forward Feature Selection

To combat high dimensionality and multicollinearity, a rigorous two-stage backward-forward selection strategy based on Ridge Regression was executed. First, **backward ranking** quantified the drop in validation R^2 upon removing each individual feature; larger drops signify greater predictive contribution. Second, **forward selection** iteratively introduced features in ranked order until validation performance peaked. This yielded an optimal subset of **21 features**. The analysis confirmed that **separation distance (*d*)** and **average PGA** are the two dominant physical determinants governing ground motion orientation correlation.

4. Methodology & Machine Learning Architectures

The curated dataset (80% train / 10% validation / 10% hold-out test) was evaluated across three core architectures:

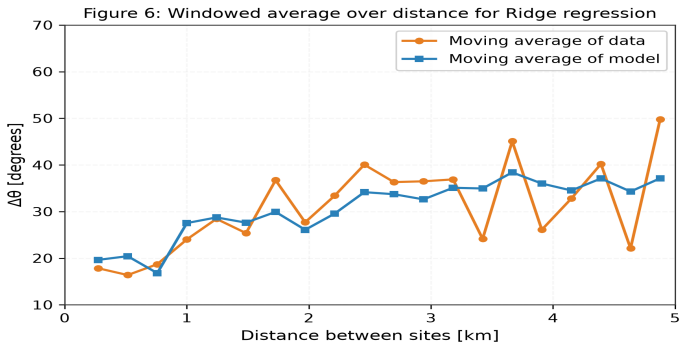
• **Benchmark Linear Regression:** Simple univariate OLS model utilizing solely separation distance d: $\Delta\theta = c_1 + c_2 d$. Serves as the baseline reflecting current spatial distance decay relations.

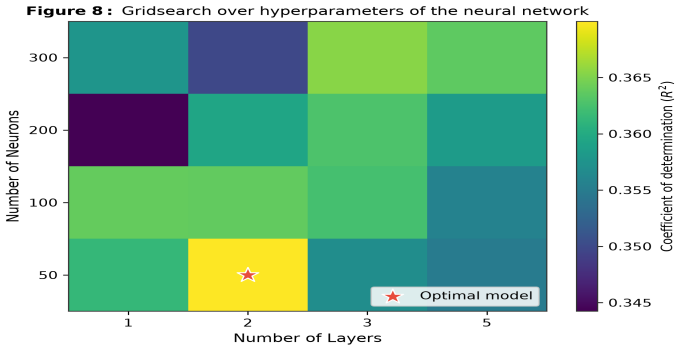
• **Regularized Ridge Regression:** Multivariate L_2-penalized linear model ($\min_\beta ||y - X\beta||_2^2 + \alpha||\beta||_2^2$) calibrated on the 21 selected features with hyperparameter α tuned on validation data (optimal $\alpha = 1.5$).

• **Deep Neural Network (MLP):** High-capacity Multi-Layer Perceptron utilizing ReLU activations and Adam optimization with L_2 weight decay ($\alpha = 0.1$). A grid search explored network depth (1 to 5 hidden layers) and neuron widths (50 to 400), selecting an optimal topology that captures complex non-linear wave interactions.

5. Experimental Results & Performance Benchmarking

Model Architecture	Inputs	Train R^2	Test R^2	Test MAE (°)	Test RMSE (°)
Benchmark (Distance Only)	1 Feature (d)	0.038	0.051	16.43°	19.95°
Ridge Regression (L2)	21 Features	0.349	0.350	13.35°	16.52°
Deep Neural Network (MLP)	21 Features	0.381	0.328	13.56°	16.79°
Gradient Boosted Trees (Bonus)	21 Features	0.751	0.325	13.71°	16.83°





Figures 3 & 4: *Left: Ridge model windowed moving average tracking empirical distance trend without overfitting. Right: Neural Network grid search contour across layer depths and neuron counts evaluating validation R^2.*

6. Key Findings & Discussion on Aleatoric Uncertainty

1. Dominance of Distance and Shaking Amplitude: As demonstrated by both the feature ranking and windowed curves, stations separated by < 1.5 km and experiencing strong PGA (> 0.15g) exhibit tight angular coherence ($\Delta\theta \approx 15°–20°$). Beyond 4 km or at low accelerations, local scattering decorrelates wave motions towards uniform randomness ($\Delta\theta \approx 45°$).

2. Linear vs Non-Linear Performance: Ridge regression drastically outperforms the baseline (R^2 jumps from 0.051 to 0.350, MAE drops from 16.4° to 13.3°). Deep neural networks and gradient boosted trees achieve comparable test errors while fitting non-linear radiation interactions.

3. Inherent Seismic Stochasticity: The maximum achievable R^2 is bounded (~0.35–0.40) due to large *aleatoric uncertainty* originating from unobservable 3D subsurface heterogeneities, crustal velocity variations, and micro-topography. While point predictions exhibit scatter, the models accurately reconstruct the physical expected mean trend across all distance and PGA bins.

7. Conclusions & Real-World Impact

This project successfully developed and validated machine learning models that predict earthquake ground motion directionality from seismological and geotechnical parameters. Replacing the traditional assumption of independent uniform orientations with the developed predictive models enables realistic multi-site ground shaking simulations for urban regions.

Future Extensions: (i) Formulating probabilistic heteroscedastic loss functions to explicitly model both conditional mean and variance $\sigma^2(\mathbf{x})$; (ii) Extending pairwise predictions to full joint multivariate spatial Gaussian random fields for simultaneous regional infrastructure resilience evaluation.